Accessibility duality: inverse-box inclusion equals diamond-image inclusion

The Open Logic Project contributors

openlogic:normal-modal-logic/completeness:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the only-if opening line has the author's slight misphrase (supposing Diamond A lies in the Diamond-image 'i.e. A in Delta') - kept verbatim.

Editorial note. The label lem:box-iff-diamond names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Completeness of maximal consistent sets (negation split). Canonical-model section standing facts: maximality places exactly one of A, not-A in Delta. Lines 1302-1303 and 1310-1311: both directions pivot on flipping membership through negation.
- Duality of box and diamond in a normal modal logic. Normality: Diamond not-A is equivalent to not Box A, used inside Gamma. Lines 1311-1313 of the reverse direction.

Lemma 0.1. *Suppose* Γ *and* Δ *are complete* Σ*-consistent. Then* $\Box^{-1}\Gamma \subseteq \Delta$ *if and only if* $\Diamond\Delta \subseteq \Gamma$.

Proof. "Only if" direction: Assume $\Box^{-1}\Gamma \subseteq \Delta$ and suppose $\Diamond A \in \Diamond\Delta$ (i.e., $A \in \Delta$). In order to show $\Diamond A \in \Gamma$, it suffices to show $\Box\neg A \notin \Gamma$, for then by maximality, $\neg\Box\neg A \in \Gamma$. Now, if $\Box\neg A \in \Gamma$ then by hypothesis $\neg A \in \Delta$, against the consistency of Δ (since $A \in \Delta$). Hence $\Box\neg A \notin \Gamma$, as required.

"If" direction: Assume $\Diamond\Delta \subseteq \Gamma$. We argue contrapositively: suppose $A \notin \Delta$ in order to show $\Box A \notin \Gamma$. If $A \notin \Delta$ then by maximality $\neg A \in \Delta$ and so by hypothesis $\Diamond\neg A \in \Gamma$. But in a normal modal logic $\Diamond\neg A$ is equivalent to $\neg\Box A$, and if the latter is in Γ, by consistency $\Box A \notin \Gamma$, as required. $\Box$